YONGGANG JIANG

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EDUCATION

Max Planck Institute for Informatics and Saarland University *Apr. 2023 – Present*
PhD in Computer Science
Advisor: Danupon Nanongkai

Max Planck Institute for Informatics and Saarland University *Oct. 2021 – Mar. 2023*
Preparatory Phase for Doctoral Study

Nanjing University *Sep. 2017 – Jul. 2021*
B.S. in Computer Science and Technology
GPA: 94.4/100, Ranking: 1/31

University of California, Berkeley *Jan. 2020 – May 2020*
Berkeley International Study Program
GPA: 4.0/4.0

HONORS & AWARDS

Distinguished Paper Award, SPAA 2025 *2025*
Outstanding Graduate, Nanjing University *2021*
National Elite Program Scholarship, Outstanding Prize (top 1) **2018, 2019, 2020**
China Collegiate Programming Contest (CCPC), Gold Medal *2018*
National Olympiad in Informatics in Provinces (NOIP), First Prize *2015*

PUBLICATIONS

***Co-authors are listed in alphabetical order.**

[1] **Reviving Thorup's Shortcut Conjecture**
Aaron Bernstein, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
In submission.

[2] **Minimum s-t Cuts with Fewer Cut Queries**
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
In submission.

[3] **Reducing Shortcut and Hopset Constructions to Shallow Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
In submission.

[4] **Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs**
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
In submission.

[5] **Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth**
Yonggang Jiang, Changki Yun.
In submission.

[6] **Shortcuts and Transitive-Closure Spanners Approximation**
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
In submission.

[7] **New Oracles and Labeling Schemes for Vertex Cut Queries**
Yonggang Jiang, Merav Parter, Asaf Petruschka.
In submission.

[8] **Perfect Simulation of Las Vegas Algorithms via Local Computation**
Xinyu Fu, Yonggang Jiang, Yitong Yin.
In submission.

[9] **Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work**
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.

[10] **Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances**
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award

[11] **Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness**
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[12] **Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations**
Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

[13] **Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights**
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.

[14] **Finding a Small Vertex Cut on Distributed Networks**
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.

[15] **Robust and Optimal Contention Resolution without Collision Detection**
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.

[16] **Tight Trade-off in Contention Resolution without Collision Detection**
Haimin Chen, Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Principles of Distributed Computing (PODC) 2021.